

MOBILE APP USER BEHAVIOR ANALYSIS & ENGAGEMENT INSIGHTS



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INTRODUCTION

Mobile applications have become an essential part of daily life, with users spending significant time on their devices.

Understanding user behavior helps in identifying usage patterns, engagement levels, and performance factors.

This project analyzes mobile device usage data to gain insights into how users interact with applications.



PROBLEM STATEMENT ↘

- Large amounts of mobile user behavior data are generated daily
- Difficulty in identifying patterns in app usage, screen time, and data consumption
- Impact of factors like age, device model, and operating system is unclear
- Challenges in understanding user engagement levels and usage habits
- Need for analysis to extract insights and improve user experience

AIM OF THE PROJECT

- To analyze mobile device usage and user behavior patterns
- To identify high usage users based on app activity and screen time
- To compare usage across different devices and operating systems
- To understand factors influencing user engagement levels
- To extract meaningful insights for improving user experience

DATASET DESCRIPTION

- The dataset contains detailed information on mobile device usage and user behavior
- Includes demographic attributes such as age and gender
- Covers device details like device model and operating system
- Contains usage metrics including app usage time, screen-on time, and session count
- Includes performance indicators such as battery drain and data usage
- Classifies users based on engagement level to understand activity patterns

DATASET

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelpAcrobat

Paste

Clipboard

Aptos Narrow

11

A⁺

A⁻

B

I

U

Font

Alignment

General

Number

Conditional Formatting

Format as Table

Cell Styles

Styles

Insert

Delete

Format

Cells

Editing

Sensitivity

Add-ins

SensitivityAdd-ins

Create a PDF

Adobe Acro...

Add-insAdobe Acro...

P1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	User ID	Device Model	Operating Sy	App Usage Time	Screen On Time	Battery Drain	Number of Apps	Data Usage	Age	Gender	User Behavior Class	Location	Engagement	Session_Count				
2	1	Google Pixel 5	Android	393	6.4	1872	67	1122	40	Male		4 Hyderabad	High	3				
3	2	OnePlus 9	Android	268	4.7	1331	42	944	47	Female		3 Bangalore	Medium	10				
4	3	Xiaomi Mi 11	Android	154	4	761	32	322	42	Male		2 Delhi	Medium	9				
5	4	Google Pixel 5	Android	239	4.8	1676	56	871	20	Male		3 Mumbai	Medium	10				
6	5	iPhone 12	iOS	187	4.3	1367	58	988	31	Female		3 Delhi	Medium	2				
7	6	Google Pixel 5	Android	99	2	940	35	564	31	Male		2 Mumbai	Low	7				
8	7	Samsung Galaxy S	Android	350	7.3	1802	66	1054	21	Female		4 Mumbai	High	1				
9	8	OnePlus 9	Android	543	11.4	2956	82	1702	31	Male		5 Bangalore	High	4				
10	9	Samsung Galaxy S	Android	340	7.7	2138	75	1053	42	Female		4 Hyderabad	High	10				
11	10	iPhone 12	iOS	424	6.6	1957	75	1301	42	Male		4 Chennai	High	6				
12	11	Google Pixel 5	Android	53	1.4	435	17	162	34	Female		1 Hyderabad	Low	7				
13	12	OnePlus 9	Android	215	5.5	1690	47	641	24	Male		3 Hyderabad	Medium	3				
14	13	OnePlus 9	Android	462	6.2	2303	65	1099	57	Female		4 Delhi	High	7				
15	14	Xiaomi Mi 11	Android	215	4.9	1662	43	857	43	Male		3 Chennai	Medium	10				
16	15	iPhone 12	iOS	189	5.4	1754	53	779	49	Female		3 Delhi	Medium	2				
17	16	Google Pixel 5	Android	503	10.4	2571	84	2025	39	Female		5 Mumbai	High	8				
18	17	OnePlus 9	Android	132	3.6	628	32	344	47	Female		2 Hyderabad	Low	10				
19	18	iPhone 12	iOS	299	5.8	1431	41	985	44	Female		3 Bangalore	Medium	2				
20	19	Google Pixel 5	Android	81	1.4	558	16	297	26	Female		1 Delhi	Low	2				
21	20	iPhone 12	iOS	577	8.5	2774	89	2192	29	Female		5 Hyderabad	High	7				
22	21	Samsung Galaxy S	Android	93	2.6	681	37	302	45	Female		2 Mumbai	Low	4				
23	22	OnePlus 9	Android	576	11.6	2803	82	1553	43	Female		5 Hyderabad	High	1				
24	23	Samsung Galaxy S	Android	423	6.5	2094	65	1372	23	Female		4 Mumbai	High	10				
25	24	Google Pixel 5	Android	292	5.6	1401	46	949	37	Female		3 Hyderabad	Medium	9				
26	25	OnePlus 9	Android	216	4	1711	59	748	58	Male		3 Hyderabad	Medium	6				
27	26	Samsung Galaxy S	Android	91	3.4	1073	38	451	52	Male		2 Chennai	Low	10				

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Raw data

Remove Duplicates & Dtype

Handling missing values

Lookup functions

Cc +

TOOLS USED

EXCEL

- Used for storing and organizing raw dataset
- Performed basic preprocessing like removing duplicates

PYTHON

- Used for data analysis and visualization (EDA)
- Identified patterns and relationships in user behavior

POWER BI

- Built interactive dashboard for data visualization
- Created KPIs and used slicers for filtering insights

SQL

- Created database and tables for structured data
- Performed queries like joins and aggregations

DATA CLEANING PROCESS

- The dataset was checked for missing or null values to ensure completeness of user behavior data
- Missing values were handled appropriately based on the type of data
- Duplicate records were identified and removed to maintain data accuracy
- Inconsistent values in columns such as device model, operating system, and engagement level were corrected
- Column names and formats were standardized for better clarity and consistency
- Data types were converted where required, such as usage time and numeric fields
- The dataset was reviewed to ensure accuracy and reliability of information
- Unnecessary or irrelevant data was removed to simplify the dataset
- The cleaned dataset was prepared for further analysis and visualization
- The final dataset was used for analysis in Python, SQL, and Power BI



EXPLORATORY DATA ANALYSIS

- Initial data exploration was performed to understand the structure and characteristics of the user behavior dataset
- Summary statistics such as mean, median, and standard deviation were used to analyze numerical variables like app usage time and screen time
- The distribution of key variables such as engagement level, device model, and gender was examined
- Trends across age and usage patterns were analyzed to understand user behavior
- User activity patterns were compared across different device models and locations
- Data was grouped using techniques such as grouping by engagement level and user behavior class
- Patterns such as high usage users and low engagement users were identified
- Relationships between variables such as usage time, screen time, and data usage were examined
- Aggregation techniques were used to summarize data, such as average usage and total users
- This analysis helped in building a strong foundation for visualization and generating user behavior insights

VISUALIZATIONS

CONTENT:

Python was used to create charts to visualize user behavior patterns and analyze relationships between variables.

CHARTS CREATED:

- Histogram – shows distribution of numerical variables such as app usage time
- Bar Chart – displays count of categorical variables like device model or gender
- Scatter Plot – shows relationship between two numerical variables
- Box Plot – compares distributions and helps identify outliers
- Heatmap – visualizes correlation between numerical variables
- Pairplot – shows relationships between multiple numerical features



DASHBOARD OVERVIEW

CONTENT:

Power BI was used to design an interactive dashboard to analyze user behavior and visualize key insights effectively.

FEATURES:

- KPI Cards – display key metrics such as total users, average usage time, and average screen time
- Bar Chart – compares user behavior across different user behavior classes
- Line Chart – shows trends in screen-on time across different age groups
- Donut Chart – represents distribution of users based on location
- Waterfall Chart – shows contribution of device models to high usage users
- Slicers – allow filtering based on engagement level and battery drain
- Interactive Dashboard – enables dynamic filtering and better data exploration



SQL ANALYSIS

CONTENT:

SQL was used to create a structured database and perform queries to analyze user behavior data.

QUERIES PERFORMED:

- Created database and tables for storing user behavior data
- Inserted dataset records into tables
- Used JOIN operations to combine data from multiple tables
- Calculated total and average values such as app usage time and screen time
- Applied GROUP BY and ORDER BY to summarize and sort data
- Identified high usage users using filtering conditions



Users with higher app usage time tend to have increased screen-on time and data usage

High engagement users show significantly higher activity compared to low engagement users

Certain device models contribute more to high usage patterns

Younger users generally exhibit higher mobile usage and engagement levels

Higher session count is directly associated with increased app usage time

Battery drain increases with higher usage and prolonged screen-on time

KEY INSIGHTS



BUSINESS RECOMMENDATIONS

- Based on the analysis of user behavior data, improving user engagement and app performance is essential. High engagement users show consistent usage patterns, so features that drive engagement should be enhanced to improve overall user experience
- Low engagement users should be targeted with personalized recommendations and improvements in app usability to increase interaction and retention
- Device-specific performance optimization should be considered, as certain device models contribute more to high usage patterns
- Battery optimization strategies should be implemented, as higher usage leads to increased battery drain, which may affect user satisfaction
- Monitoring user sessions and usage patterns can help identify peak usage times and optimize application performance accordingly

CONCLUSION

The mobile app user behavior analysis provides a clear understanding of how users interact with mobile devices and applications. By analyzing key metrics such as app usage time, screen-on time, session count, and engagement levels, important patterns in user behavior were identified

The project involved multiple stages including data cleaning, exploratory data analysis, SQL querying, and dashboard visualization. Tools such as Excel, Python, SQL, and Power BI were effectively used to process and analyze the dataset

Visualizations and analysis helped in identifying high usage users, engagement patterns, and relationships between different variables. These insights are useful for improving user experience, optimizing application performance, and supporting data-driven decision-making

Overall, the project demonstrates how data analysis techniques can be used to understand user behavior and generate meaningful insights for better application development and performance





THANK YOU